II Semester			
S. No	**Course Code**	**Course Title**	**Credit Hours**
1.	**HORT-121**	Post Harvest Processing Technology*	2(0+2)
2.	**PPATH-121**	Production Technology of Bio-Agent*	2(0+2)
3.	**PPATH-122**	Fundamentals of Plant Pathology	3(2+1)
4.	**AGEXT-121**	Personality Development	2(1+1)
5.	**ENTO-121**	Fundamentals of Entomology	3(2+1)
6.	**ESDM-121**	Environmental Studies & Disaster Management	3(2+1)
7.	**LPM-121**	Livestock and Poultry Management	2(1+1)
8.	**SSAC-121**	Soil Fertility Management	3(2+1)
9.	**NCC-II/ NSS-II /PHED-II**	NSS/NCC/Physical Education &Yoga Practices	1(0+1)
Total Credit			**21 (10+11)**
***Skill Enhancement Courses**			

HORT-121 POSTHARVEST PROCESSING TECHNOLOGY 2(0+2)

Objectives

1. To provide knowledge on postharvest processing technology of various horticultural produce.
1. To provide knowledge on value addition of horticultural crops.
2. To provide knowledge on post-harvest management of horticultural crops.

Practical

Practice in judging the maturity of various horticultural produce, Determination of physiological loss in weight and quality. Grading of horticultural produce, post-harvest treatment of horticultural crops. Packaging studies in fruits, vegetables, plantation crops and cut flowers by using different packaging materials and Storage methods. Post-harvest disorders in horticultural produce. Equipment used in post-harvest technology processing units. Canning of fruits and vegetables. Preparation of squash, RTS, cordial, syrup, jam, jelly and marmalade. Preparation of candies, preserves, chutneys, sauces and pickles (hot and sweet). Dehydration of fruits and vegetables. Visit to markets/processing unit/packaging houses/cold storage units.

Lecture schedule: Practical

S. No.	Topic	No. of lectures
1.	Practice in judging the maturity of various horticultural produce	2
2.	Determination of physiological loss in weight and quality	2
3.	Grading of horticultural produce	1
4.	Postharvest treatment of horticultural crops	2
5.	Packaging studies in fruits, vegetables, plantation crops and cut flowers by using different packaging materials	3
6.	Storage methods	3
7.	Postharvest disorders in horticultural produce	3
8.	Equipment used in food processing units	2
9.	Canning of fruits and vegetables	2
10.	Preparation of squash, RTS, cordial, syrup	2
11.	Jam, jelly, marmalade	2
12.	Preparation of candies, preserves	2
13.	Chutneys, sauces, pickles (hot and sweet)	2
14.	Dehydration of fruits and vegetables	2
15.	Visit to markets/processing unit/packaging houses/cold storage units	2
		32

Suggested Readings

1. Kader, A. A. 2002. Postharvest Technology of Horticultural Crops. UCUCANR Publications. 535p.
2. Ranganna, S. 1986. Handbook of Analysis and Quality Control for Fruit and VegetableProducts. Tata Mc. Graw Hill Publishing Company, New Delhi, 1112p.
3. Saraswathy, S., Preeti, J. L., Balasubramanyan, S., Suresh, J., Revathy, N. and Natarajan, S. 2008. Postharvest management of horticultural crops. AGRIBIOS (India).
4. Sharma, S. K. 2010. Post-harvest management and processing of fruits and vegetables-Instant Notes. New India Publishing Agency. New Delhi.
5. Srivastava, R. P. and Kumar, S. 2007. Fruits and vegetable Preservation: Principles and Practice. International Book Distributing Company, Lucknow.
6. Sudheer, K. P. and Indira, V. 2007. Postharvest Technology of Horticultural Crops. New India. Publ. Agency.

7. Verma, L. R. and Joshi, V. K. 2000. Postharvest technology of fruits and vegetables General concepts and principles. Vol I & II. Indus Publishing Co., New Delhi.
8. Yadav, P. K. and Singh, J. 2000. Phalotpadan and Prasanskaran Published.in 2000 from Agro-bios (India), Jodhpur., ISBN 81-7754-086-6

PPATH-121 PRODUCTION TECHNOLOGY OF BIOAGENTS 2 (0+2)

Objectives

1. To study principle sand application of eco friendly and sustainable management strategies of plant diseases.

Practical:

Concept of biological control, definitions, importance, principles of plant disease management with bioagents, history of biological control, merit sand demerits of biological control. Types of biological interactions, competition: mycoparasitism, exploitation for hypovirulence, rhizosphere colonization, competitive saprophytic ability, antibiosis, induced resistance, mycorrhizal associations, operational mechanism sand it relevance in biological control. Factors governing biological control, role of physical environment, agro ecosystem, operational mechanisms and cultural practices in biological control of pathogens, pathogens and antagonists and their relationship, bio control agents, comparative approaches to biological control of plant pathogens by resident and introduced antagonists, control of soil-borne and foliar diseases. Compatibility of bioagents with agrochemicals and other antagonistic microbes. Commercial production of antagonists, their delivery systems, application and monitoring, biological control in IDM, IPM and organic farming system, biopesticides available in market. Quality control system of biocontrol agents. Isolation, characterization and maintenance of antagonists, methods of study of antagonism and antibiosis, application of antagonists against pathogen *in vitro and in vivo* conditions. Preparation of different formulations of selected bioagents and their mass production. Quality parameters of biocontrol agents. One week exposure visit to commercial bio control agents production unit.

Lecture Schedule: Practical

S. No.	Topic	No of Lectures
1.	Concept of definitions, importance of biological control	1
2.	Principles of plant disease management with bioagents, history of biological control, merits and demerits of biological control.	2

3.	Types of biological interactions, competition: mycoparasitism, exploitation for hypovirulence, rhizosphere colonization	1
4.	Competitive saprophytic ability and antibiosis	2
5.	Induced resistance and mycorrhizal associations	1
6.	Operational mechanisms and it srelevance in biological control.	2
7.	Factorsgoverningbiologicalcontrol,roleofphysicalenvironment,agroecosystem,operationalmechanisms and cultural practices in biological control of pathogens	2
8.	Pathogens and antagonists and their relationship, biocontrol agents, comparative approaches to biological control of plant pathogens by resident and introduced antagonists	2
9.	Control of soil-borne and foliar diseases.	2
10.	Compatibility of bioagents with agro chemicals and other antagonistic microbes	2
11.	Commercial production of antagonists, their delivery systems, application and monitoring, biological control in IDM,	2
12.	IPM and organic farming	1
13.	Biopesticides available in market.	2
14.	System Quality control system of biocontrol agents.	2
15.	Isolation, characterization and maintenance of antagonists	2
16.	Methods of study of antagonism and antibiosis	1
17.	Application of antagonists against pathogen *in vitro and in vivo* conditions.	2
18.	Preparation of different formulations of selected bioagents and their mass production.	1
19.	Quality parameters of biocontrol agents.	1
20.	One week exposure visit to commercial biocontrol agents production unit	1

Suggested Readings:

CampbellR. 1989. *Biological Control of Microbial Plant Pathogens*. Cambridge Univ. Press, Cambridge.

Cook RJ & Baker KF. 1983. *Nature and Practice of Biological Control of Plant Pathogens*. APS, St. Paul, Minnesota.

FokkemmaMJ.1986.*MicrobiologyofthePhyllosphere*.CambridgeUniv. Press, Cambridge.GnanamanickamSS(Eds).2002.*BiologicalControlofCropDiseases*.CRCPress, Florida.

Heikki MT & Hokkanen James M (Eds.). 1996. *Biological Control - Benefits and Risks*.

Cambridge Univ. Press, Cambridge.

Mukerji KG,TewariJP,AroraDK&SaxenaG.1992.*RecentDevelopmentsin Biocontrol of Plant Diseases*. Aditya Books, New Delhi.

PPATH-122 FUNDAMENTALS OF PLANT PATHOLOGY 3(2+1)

Objectives

1. To get acquainted with the role of different microorganisms in the development of plant disease
2. To get general concepts and classification of plant diseases
3. To get knowledge of general characteristics of fungi, bacteria, virus, and other microorganisms causing plant diseases
4. To acquaint the students with reproduction in fungi, and bacteria, causing plant diseases
5. To get acquainted with various plant disease management principles and practices

Theory:

Introduction to Plant Pathology: Concept of disease in plants, Importance of Plant disease, Scope & Objective of Plant pathology, History of Plant Pathology with special references to India; Terms & concepts in Plant Pathology Pathogenesis. Causes and classification of plant diseases, diseases and symptoms due to biotic/ abiotic causes. Parasitism and pathogenesis; Development of disease in plants: Disease Triangle, Disease cycle. Fungi: general Characters, Somatic structures, Types of fungi thalli, fungi tissues, modification of thallus and their morphology, reproduction (Sexual & Asexual) Nomenclature, Binomial system of nomenclature, Classification of fungi (key to Domain to Phylum). Bacteria and mollicutes: general morphological characters. Reproduction and classification of plant pathogenic bacteria. Important plant pathogens: Mollicutes; Flagellant protozoa; FVB; Green algae and parasitic higher plants, prions & viroids Viruses : Nature , Structure and Transmission Role of enzymes and toxins in disease development. Defense mechanism in plants. Principles of Plant disease management. Disease management with chemicals, Host resistance, cultural and biological method. Principals of IDM & modules for What, Rice, Sugarcane, Cotton, groundnut, citrus and chickpea.

Lecture Schedule: Theory

S.No	Topic	No. of lectures
1.	Introduction to Plant Pathology: Concept of disease in plants, Importance of	02

	Plant disease, Scope & Objective of Plant pathology	
2.	History of Plant Pathology with special references to India; Terms & concepts in Plant Pathology	02
3.	Pathogenesis. Causes and classification of plant diseases, diseases and symptoms due to biotic/ abiotic causes.	02
4.	Parasitism and pathogenesis; Development of disease in plants: Disease Triangle, Disease cycle.	01
5.	Fungi: general Characters, Somatic structures, Types of fungi thalli, fungi tissues, modification of thallus and their morphology, reproduction (Sexual & Asexual)	03
6.	Nomenclature, Binomial system of nomenclature, Classification of fungi (key to Domain to Phylum).	03
7.	Bacteria and mollicutes: general morphological characters	02
8.	Reproduction and classification of plant pathogenic bacteria.	02
9.	Important plant pathogens: Mollicutes; Flagellant protozoa; FVB; Green algae and parasitic higher plants, prions &viroids	03
10.	Viruses:Nature , Structure and Transmission	02
11.	Role of enzymes and toxins in disease development. Defense mechanism in plants.	02
12.	Principles of Plant disease management.	02
13.	Disease management with chemicals, Host resistance, cultural and biological method	03
14.	Principes of IDM & modules for What, Rice, Sugarcane, Cotton, groundnut, citrus and chickpea.	03

Practical

Study of the microscope; Acquaintance with laboratory material and equipment; Study of different plant disease symptoms; Microscopic examination of general structure of fungi; Simple staining of bacteria: Direct and indirect staining, Gram staining of bacteria; Microscopic examination of fungal disease specimen; Microscopic examination of bacterial disease specimen; Preparation of culture media; Isolation of plant pathogens: Fungi, bacteria and viruses; Purification of plant pathogens; Study on plant disease diagnosis: Koch’s Postulates, Characteristics, formulation, methods of application and calculation on fungicides.

Lecture Schedule: Practical

S.No	Topic	No. of lectures
1.	Study of the microscope; Acquaintance with laboratory material and equipment	02
2.	Study of different plant disease symptoms	01
3.	Microscopic examination of general structure of fungi	01
4.	Simple staining of bacteria: Direct and indirect staining, Gram staining of bacteria	02
5.	Microscopic examination of fungal diseased specimen	01
6.	Microscopic examination of bacterial diseased specimen	01
7.	Preparation of culture media	01
8.	Isolation of plant pathogens: Fungi, bacteria and viruses	02
9.	Purification of plant pathogens	01
10.	Study on plant disease diagnosis	01
11.	Koch's Postulates	01
12.	Characteristics, formulation, methods of application and calculation on fungicides.	02

Suggested readings

1. Agrios, G.N. 2010. Plant Pathology. Acad. Press.

2. Alexopoulos, Mims and Blackwel. Introductory Mycology.

3. Dhingra, O.D. and Sinclair, J.B. 1986. Basic Plant Pathology Methods. CRC Press, London,Tokyo.

4. Gibbs, A. and Harrison, B. 1976. Plant Virology - The Principles. Edward Arnold, London

5. Goto, M. 1990. Fundamentals of Plant Bacteriology. Academic Press, New York.

6. Hull R. 2002. Mathew's Plant Virology. 4th edn. Academic Press, New York.

7. Kamat, M. N. Introductory Plant Pathology. Prakash Pub, Jaipur.

AGEXT- 121 PERSONALITY DEVELOPMENT 2 (1+1)

Objective

To make students realize their potential strengths, cultivate their inter-personal skills and improve employability.

Theory

Personality Definition, Nature of personality, theories of personality and its types. The humanistic approach - Maslow’s self-actualization theory, shaping of personality, determinants of personality, Myers-Briggs Typology Indicator, Locus of control and performance, Type A and Type B Behaviours, personality and Organizational Behaviour. Foundations of individual behavior and factors influencing individual behavior, Modelsof individual behavior, Perception and attributes and factors affecting perception, Attribution theory and case studies on Perception and Attribution. Learning: Meaning and definition, theories and principles of learning, Learning and organizational behavior, Learning and training, learning feedback. Attitude and values, Intelligence- types of Intelligence, theories of intelligence, measurements of intelligence, factors influencing intelligence, intelligence and Organizational behavior, emotional intelligence. Motivation- theories and principles, Teamwork and group dynamics.

Lecture Schedule

S.N.	Topic	No. of Lectures
1.	Personality: Definition, Nature of Personality, Theories of Personality and Its Types	2
2.	The Humanistic Approach-Maslow’s Self-Actualization Theory, Shaping of Personality, Determinants of Personality	1
3.	Myers-Briggs Typology Indicator, Locus of Control and Performance, Type A and Type B Behaviours	1
4.	Personality and Organizational Behaviour, Foundations of Individual Behavior and Factors Influencing Individual Behavior, Models of Individual Behavior	2
5.	Perception and Attributes and Factors Affecting Perception, Attribution Theory and Case Studies on Perception and Attribution	2
6.	Learning: Meaning and Definition, Theories and Principles of Learning, Learning and Organizational Behavior, Learning and Training, Learning Feedback	2
7.	Attitude and Values	1
8.	Intelligence: Types of Intelligence, Theories of Intelligence, Measurements of Intelligence, Factors Influencing Intelligence, Intelligence and Organizational Behavior	2

9.	Emotional Intelligence	1
10.	Motivation: Theories and Principles	1
11.	Teamwork and Group Dynamics	1

Practical

MBTI personality analysis, Learning Styles and Strategies, Motivational needs, Firo-B, Inter personal Communication, Teamwork and team building, Group Dynamics, Win-win game, Conflict Management, Leadership styles, Case studies on Personality and Organizational Behavior.

S.N.	**Topic**	**No. of Lectures**
1.	MBTI Personality Analysis	2
2.	Learning Styles and Strategies	2
3.	Motivational Needs, Firo-B	2
4.	Interpersonal Communication	2
5.	Teamwork and Team Building	1
6.	Group Dynamics	2
7.	Win-Win Game	1
8.	Conflict Management	1
9.	Leadership Styles	1
10.	Case Studies on Personality and Organizational Behavior	2

Suggested reading

1. Andrews, Sudhir. 1988. How to Succeed at Interviews. 21st (rep.) New Delhi. Tata McGraw-Hill.

2. Heller, Robert. 2002. Effective Leadership. Essential Manager series. Dk Publishing.

3. Hindle, Tim. 2003. Reducing Stress. Essential Manager series. Dk Publishing.

4. Lucas, Stephen. 2001. Art of Public Speaking. New Delhi. Tata - Mc-Graw Hill.

5. Mile, D.J. 2004. Power of Positive Thinking. Delhi. Rohan Book Company.

6. Pravesh Kumar. 2005. All about Self- Motivation. New Delhi. Goodwill Publishing House.

7. Smith, B. 2004. Body Language. Delhi: Rohan Book Company.

8. Shaffer, D. R. 2009. Social and Personality Development (6th Edition). Belmont, CA: Wadswor.

ENTO- 121 FUNDAMENTALS OF ENTOMOLOGY 3(2+1)

Objectives

1. To know the history of entomology, classification of insects and their relationship with other arthropods
2. To study thevariousmorphologicalcharactersofclassinsectandtheirimportancefor classification of insects
3. Togetanideaaboutthedifferentphysiologicalsystemsofinsectsandtheirrolesin growth and development and communications of insects
4. To study the characteristics of commonly observed insect orders and their economically important families

Theory

History of Entomology in India. Major points related to dominance of Insects in Animal kingdom. Classification of phylum Arthropoda up to classes. Relationship of class Insects with other classes of Arthropoda. Morphology: Structure and functions of insect cuticle and molting. Body segmentation. Structure of head, thorax and abdomen. Structure and modifications ofinsect antennae, mouth parts, legs, Wing venation, modifications and wing Structure and functions of digestive, circulatory, excretory, respiratory, nervous, secretary (Endocrine) and reproductive system, in insects. Types of reproduction in insects. Systematics: Taxonomy - importance, history and development and binomial nomenclature. Definitions of Biotype, Sub- species, Species, Genus, Family and Order. Classification of class Insecta up to Orders, basic groups of present day insects with special emphasis to orders and families of Agricultural importance like Orthoptera: Acrididae, Tettigoniidae, ryllidae, Gryllotalpidae; Chrysopidae; Lepidoptera: Papiloinidae, Noctuidae, Sphingidae, pyralidae, Gelechiidae, Arctiidae, Saturnidae, Bombycidae; Coleoptera: Coccinellidae, Chrysomelidae, Cerambycidae, Curculionidae, Bruchidae, Scarabaeidae; Hymenoptera: Tenthridinidae, Apidae. Trichogrammatidae, Ichneumonidae, Braconidae, Chalcididae; Diptera: Cecidomyiidae, Tachinidae, Agromyziidae, Culicidae, Muscidae, Tephritidae.

Lecture Schedule: Theory

S.N.	Topic	No. of lectures

1.	History of Entomology in India.	1
2.	Major points related to dominance of Insect in animal kingdom.	1
3.	Classification of phylum Arthropoda upto classes.	1
4.	Structure and functions of insect cuticle and molting.	1
5.	Morphology of grass hopper: Body segmentation-structure of head, Thorax and abdomen.	2
6.	Structure and modifications of insect antennae.	1
7.	Structure and modifications of insect mouthparts.	3
8.	Structure and modifications of insect leg.	1
9.	Wing venation, modifications and wing coupling apparatus.	1
10.	Structure of genital organs and sensory organs(simple and compound eyes, chemoreceptor).	2
11.	Metamorphosis ninsects, types of larvae and pupae.	1
12.	Structure and functions of digestive system.	1
13.	Structure and functions of circulatory and excretory system.	2
14.	Structure and functions of respiratory system.	1
15.	Structure and functions of nervous system.	1
16.	Structure and functions of secretory (endocrine) system	1
17.	Structure and functions of reproductive system and types of Reproduction in insects.	2
18.	**Taxonomy**-importance,historyanddevelopmentandbinomialnomenclature.DefinitionsofBiotype,Sub-species,Species,Genus, Family and Order.	2
19.	Orthoptera: Acrididae, Gryllidae; Dictyoptera: Mantidae, Blattidae; Odonata;Isoptera:Termitidae.Thysanoptera: Thripidae.	2
20.	Hemiptera: Pentatomidae, Coreidae, Pyrrhocoridae, Lygaeidae, Cicadellidae, Delphacidae, Aphididae, Coccidae, Lophophidae, Aleurodidae, Pseudococcidae.	1
21.	Lepidoptera:Pieridae,Papiloinidae,Noctuidae,Sphingidae,Pyralidae,	1
	Gelechiidae, Arctiidae,Bombycidae.	
22.	Coleoptera:Coccinellidae,Galerucidae,Cerambycidae,Curculionidae, Bruchidae..	1

23.	Hymenoptera: Tenthridinidae, Apidae, Trichogrammatidae, lchneumonidae, Braconidae, Chalcididae.	1
24.	Diptera: Cecidomyiidae, Tachinidae, Agromyziidae, Culicidae, Muscidae,Tephritidae;Neuroptera: Chrysopidae.	1

Practical

Methods of collection and preservation of insects including immature stages; External features of Grasshopper/Blister beetle; Types of insect antennae, mouthparts and legs; Wing venation,typesofwingsandwingcouplingapparatus.Typesofinsectlarvaeandpupae;

Dissection of digestive system in insects (Grasshopper); Study of characters of orders Orthoptera, Dictyoptera, Odonata, Isoptera, Thysanoptera, Hemiptera, Lepidoptera, Neuroptera, Coleoptera,Hymenoptera, Diptera and their families of agricultural importance.

Lecture Schedule: Practical

S.N.	Topic	No. of lectures
1.	Methods of collection and preservation of insects including immature stages.	1
2.	External features of Grasshopper/Cockroach.	1
3.	Types of insect antennae, mouth parts and legs.	4
4.	Wing venation, types of wings and wing coupling apparatus.	1
5.	Dissection of digestive system in insects(Grasshopper/Cockroach)	1
6.	Study of characters of orders Orthoptera, Dictyoptera with their families.	1
7.	Study of characters of orders Odonata, Isoptera,Thysanoptera with Their families.	1
8.	Study of characters of order Hemiptera with its families.	1
9.	Study of characters of order Lepidoptera with its families.	1
10.	Study of characters of order Coleoptera with its families.	1
11.	Study of characters of order Diptera with its families.	1
12.	Study of characters of orders Hymenoptera and Neuroptera with their families.	1

References:

1. Fundamentalsof Ecology-Eugene. P.Odum andGrayW. Barrett
2. Imm'sGeneralTextbookofEntomology—O.W.RechardsandR.G. Davies
3. IntroductiontothestudyofInsects-D.J.Borrorand DeLong's

ESDM-121 ENVIRONMENTAL STUDIES AND DISASTER MANAGEMENT 3 (2+1)

Objective

- Toexposeandacquireknowledgeontheenvironmentandtogainthestate-of-the-art-skill and expertise on management of disasters

Theory

Introduction to Environment-

Environmental studies: Definition, scope and importance- Multidisciplinary nature of environmental studies-Segments of Environment-Spheres of Earth -Lithosphere- Hydrosphere- Atmosphere-Different layers of atmosphere. Natural Resources:

Classification - Forest resources. Water resources. Mineral resources Food resources. Energy resources. Land resources. Soil resources. Ecosystems: Concept of an ecosystem - Structure and function of an ecosystem - Energy flow in the ecosystem. Types of ecosystem. Biodiversity and its conservation: Introduction, definition, types. Biogeographical classification of India. Importance and Value of biodiversity. Biodiversity hot spots. Threats and Conservation of biodiversity.

EnvironmentalPollution: Definition,cause,effectsandcontrolmeasuresof:a.Airpollution.

b. Water pollution. c. Soil pollution. d. Marine pollution. e. Noise pollution. f. Thermal pollution g Light pollution. Solid Waste Management: Classification of solid wastes and management methods, Composting, Incineration, Pyrolysis, Biogas production, Causes, effects and control measures of urban and industrial wastes. Social Issues and the Environment: Urban problems related to energy. Water conservation, rain water harvesting, watershed management. Environmental ethics: Issues and possible solutions, climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Environment Protection Act. Air (Prevention and Control of Pollution) Act. Water (Prevention and control of Pollution) Act. Wildlife Protection Act. Forest Conservation Act. Human Population and the Environment: Environment and human health: Human Rights, Value Education. Women and Child Welfare. Role of Information Technology in Environment and human health.

Disaster management: Disaster definition - Types - Natural Disasters - Floods, drought, cyclone, earthquakes, landslides, avalanches, volcanic eruptions, Heat and cold waves. Man Made Disasters: Nuclear disasters, chemical disasters, biological disasters, building fire, coal fire, forest fire, oil fire, road accidents, rail accidents, air accidents, sea accidents. International and National strategy for disaster reduction. Concept of disaster management, national disaster management framework; financial arrangements; role of NGOs, community-based organizations and media in disaster management. Central, state, district and local administration in disaster control; Armed forces in disaster response; Police and other organizations in disaster management.

LectureSchedule:Theory

S. N.	Topic	No. of lectures
1.	Environmental studies: Definition, scope and importance. Multidisciplinary Nature of environmental studies Definition, scope and importance.	1
	Segments of Environment-Spheres of Earth-Lithosphere-Hydrosphere-Atmosphere-Different layers of atmosphere.	1
2.	Natural Resources: Renewable and non-renewable resources, Natural Resources and associated problems.	1
3.	a)Forestresources:Useandover-exploitation,deforestation,casestudies. Timberextraction,mining,damsandtheireffectsonforestandtribalpeople.	1
4.	b)Water resources: Use and over-utilization of surface and ground water, floods, drought, conflict sover water,dams- benefitsand problems.	1
5.	c)Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies.	1
6.	d) Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems ,water logging, salinity, case studies.	1
7.	e)Energy resources: Growing energy needs, renewable and nonrenewable energy sources, use of alternate energy sources. Case studies.	1
8.	f)Land resources: Landasare source, landed gradation, maninduced landslides, soil erosion and desertification.	1
9.	Roleofanindividualinconservationofnaturalresources.Equitableuseof Resources for sustainable lifestyles.	1
10.	Ecosystems:Conceptofanecosystem,Structureandfunction,Energyflowin the ecosystem	1
11.	Biodiversityanditsconservation:-Introduction,definition,genetic,species& Ecosystem diversity and bio geographical classification of India.	2
12.	tanceandualue of biodiversity: Biodiversity hot spots,threat sand conservation of biodiversity.	1
13	Environmental Pollution: definition, cause, effects and control measures of: a. Airpollutionb.Waterpollutionc.Soilpollutiond.Marinepollutione.Noise pollutionf. Thermal pollutiong. Light pollution.	2
14.	Solid Waste Management: Classification of solid wastes and management methods,Composting,Incineration,Pyrolysis,Biogasproduction,Causes, Effect sand control measures of urban and industrial wastes.	2

15.	SocialIssuesandtheEnvironment:Urbanproblemsrelatedtoenergy.Water conservation, rainwater harvesting, watershed management	1
16.	Environmentalethics:Issuesandpossiblesolutions,climatechange,global warming, acidrain, ozonelayerdepletion,nuclearaccidentsandholocaust.	1
17	EnvironmentProtectionAct.Air(PreventionandControlofPollution)Act. Water(Prevention and control of Pollution)Act. Wild life Protection Act. Forest Conservation Act.	2
18	Human Population and the Environment: Environment and human health: Human Rights, Value Education. Women and Child Welfare. Role of Information Technology in Environment and human health.	2
19.	Disasterdefinition-Types-NaturalDisasters-Floods,drought,cyclone, earthquakes, landslides, avalanches, volcanic eruptions, Heat and cold waves.	3
20	ManMadeDisasters:Nucleardisasters,chemicaldisasters,biologicaldisasters,buildingfire,coalfire,forestfire,oilfire,roadaccidents,rail accidents, airaccidents, seaaccidents.	2
21.	International and National strategy for disaster reduction. Concept of disaster management, national disaster management framework; financial arrangements; role of NGOs, community-based organizations and media in disaster management. Central, state, district and local administration indisaster control; Armed forces in disaster response; Police and other organizations in disaster management.	3

Practical

Visit to a local area to document environmental assets river/forest/grassland/hill/mountain. Energy: Biogas production from organic wastes. Visit to wind mill / hydro power / solar power generation units. Biodiversity assessment in farming system. Floral and faunal diversity assessment in polluted and un polluted system. Visit to local polluted site - Urban/Rural/Industrial/Agricultural to study of common plants, insect sand birds. Environmental sampling and preservation. Water quality analysis: pH, EC and TDS. Estimation of Acidity, Alkalinity. Estimation of water hardness. Estimation of DO and BOD in water samples. Estimation of COD in water samples. Enumeration of *E. coli* in water sample. Assessment of Suspended Particulate Matter(SPM).Study of simple ecosystem –Visit to pond/river/hills. Visit to areas affected by natural disaster.

LectureSchedule: Practical

S. N.	Topic	No. of lectures
1	Visit to a local area to document environmental assets river/ forest/grassland/hill/mountain.	1
2	Energy: Biogasproduction from organic wastes. Visit to windmill/hydro power/solar power generation units.	2
3	Biodiversity assessment in farming system. Floral and faunal diversity Assessment in polluted and unpolluted system.	2
4	Visittolocalpollutedsite-Urban/Rural/Industrial/Agriculturaltostudyof Common plants, insect sand birds.	2
5	Environmental sampling and preservation.	2
6	Water quality analysis: pH, EC and TDS. Estimation of Acidity, Alkalinity. Estimation of water hardness. Estimation of DO and BOD in water samples. Estimation of COD in water samples.	2
7	Enumerationof*E.coli*inwatersample.AssessmentofSuspendedParticulate Matter(SPM)..	2
8	Study of simple ecosystem– Visit to pond/river/hills.	1
9	Visit to areas affected by natural disaster	

SuggestedReadings

1. De,A.K.2010.Environmentalchemistry.PublishedbyNewAgeInternational Publishers, New Delhi. ISBN:13–978 81 224 2617 5. 384 pp
2. DharChakrabarti,P.G.2011.Disastermanagement- India’sriskmanagementpolicy
3. frameworksandkeychallenges.PublishedbyCentreforSocialMarkets(India), Bangalore. 36 pp.
4. Erach Bharucha, Text book for Environmental studies. University Grants Commission, New Delhi.
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LPM-121 LIVESTOCK AND POULTRY MANAGEMENT 2(1+1)

Objectives:

1. Provide basic knowledge to the students about scientific livestock and poultry rearing practices.
2. Entrepreneurship development through livestock/poultry and agriculture integrated farming system.

Theory

Role of livestock in the national economy. Reproduction in farm animals and poultry. Housing principles, space requirements for different species of livestock and poultry. Management of calves, growing heifers and milch animals. Management of sheep, goat, camel and swine. Incubation, hatching and brooding. Management of growers and layers. Important Indian and exotic breeds of cattle, buffalo, sheep, goat, swine, camel and poultry. Improvement off arm animals and poultry. Digestive system of livestock and poultry. Classification of feed and fodder. Nutrients and their functions. Feed ingredients for ration for live stock and poultry. Feed supplements and feed additives. Feeding of livestock and poultry. Different diseases of livestock and poultry, their prevention and control.

Lecture Schedule: Theory

S. No.	Topic	No. of Lecture
1.	Role of livestock & poultry in the national economy	1
2.	Reproduction in farm animals and poultry	1
3.	Housing principles, space requirements for different species of livestock and poultry	2
4.	Management of calves, growing heifers and milch animals.	1
5.	Management of sheep, goat, camel and swine	1
6.	Incubation, hatching and brooding. Management of growers and layers	1
7.	Important Indian and exotic breeds of cattle, buffalo, sheep, goat, swine, camel and poultry	2
8.	Improvement of farm animals and poultry	1
9.	Digestive system of livestock and poultry.	1
10.	Classification of feed and fodder.	1
11.	Nutrients and their functions. Feed ingredients for ration for livestock and poultry.	1
12.	Feed supplements and feed additives. Feeding of livestock and poultry.	1
13.	Different diseases of livestock and poultry, their prevention and control.	2

	Total theory classes	16

Practical

External body parts of cattle, buffalo, sheep, goat, swine, camel and poultry. Handling and restraining of livestock& poultry. Identification methods of farm animals and poultry. Dentition of livestock. Visit to livestock and poultry farms. Daily routine farm operations and farm records of livestock and poultry farms. Judging of cattle, buffalo and poultry. Culling of livestock and poultry. Planning and layout of housing for different types of livestock& poultry. Computation and formulation of ration for livestock and poultry. Proximate principles of feed. Clean milk production, milking methods. Hatchery management and poultry farm equipment's. Management of chicks, growers and layers. Debeaking, dusting and vaccination. Economics of cattle, buffalo, sheep, goat, swine and poultry production.

Lecture Schedule:Practical

S. No.	Topic	No. of Lectures
17.	External body parts of cattle, buffalo, sheep, goat, swine, camel and poultry.	1
18.	Handling, restraining and Identification methods of farm animals and poultry and Dentition of livestock.	2
19.	Visit to livestock and poultry farms. Daily routine farm operations and farm records of livestock and poultry farms.	2
20.	Judging of cattle, buffalo and poultry. Culling of livestock and poultry.	1
21.	Planning and layout of housing for different types of livestock& poultry.	2
22.	Computation and formulation of ration for livestock and poultry.	1
23.	Proximate principles of feed.	1
24.	Clean milk production, milking methods.	1
25.	Hatchery management and poultry farm equipment. Management of chicks, growers and layers.	2
26.	Debeaking, dusting and vaccination.	1
27.	Economics of cattle, buffalo, sheep, goat, swine and poultry production.	2
	Total practical classes	**16**

Suggested Readings:

1. A Textbook of Animal Husbandry by G. C Banerjee (8th Edition)

2. Livestock Production Management by N.S.R. Sastry and C.K. Thomas (2021)

3. Handbook of Animal Husbandry by ICAR (Latest)

4. Goat, Sheep & Pig Production Management by Jagdish Prasad (2018)

5. Textbook of Commercial Poultry production and Hatchery Management by Dr. M. Murgan (2019)

SSAC-121 SOIL FERTILITY MANAGEMENT 3 (2+1)

Objective

To provide a comprehensive knowledge of soil fertility, plant nutrition, fertilizers, and nutrient management

Theory

History of soil fertility and plant nutrition. criteria of essentiality. Role, deficiency and toxicity symptoms of essential plant nutrients, Mechanisms of nutrient transport to plants, factors affecting nutrient availability to plants. Chemistry of macro and micronutrients. Soil fertility evaluation, Soil testing. Critical levels of different nutrients in soil. Forms of nutrients in soil, plant analysis, rapid plant tissue tests. Indicator plants. Introduction and importance of manures and fertilizers. Fertilizer recommendation approaches.

Integrated nutrient management. Chemical fertilizers: classification, composition and properties of major fertilizers, secondary and micronutrient fertilizers, Complex fertilizers, Customized fertilizers, water soluble fertilizers nano fertilizers Soil amendments, Fertilizer Storage, Fertilizer Control Order. Methods of fertilizer recommendations to crops. Factor influencing nutrient use efficiency (NUE), methods of application under rainfed and irrigated conditions. STCR/RTNM/IPNS, Carbon sequestration and Carbon Trading, Preparation and properties of major manures (FYM, Compost, Vermicompost, Green manuring, Oilcakes).

S.No.	Topic	No. of lectures
1.	History of soil fertility and plant nutrition	1
2.	Criteria of essentiality	1
3.	Role, deficiency and toxicity symptoms of essential plant nutrients,	3
4.	Mechanisms of nutrient transport to plants	2
5.	Factors affecting nutrient availability to plants.	2
6.	Chemistry of macro and micronutrients	2
7.	Soil fertility evaluation, Soil testing.	2
8.	Critical levels of different nutrients in soil	2
9.	Forms of nutrients in soil, plant analysis, rapid plant tissue tests.	2
10.	Indicator plants	1
11.	Introduction and importance of manures and fertilizers	1

12.	Fertilizer recommendation approaches	1
13.	Integrated nutrient management	1
14.	Chemical fertilizers: classification, composition and properties of major fertilizers, secondary and micronutrient fertilizers, Complex fertilizers, Customized fertilizers, water soluble fertilizers nano fertilizers	3
15.	Soil amendments, Fertilizer Storage, Fertilizer Control Order.	1
16.	Methods of fertilizer recommendations to crops.	1
17.	Factor influencing nutrient use efficiency (NUE), methods of application under rainfed and irrigated conditions	1
18.	STCR/RTNM/ IPNS	2
19.	Carbon sequestration and Carbon Trading	1
20.	Preparation and properties of major manures (FYM, Compost, Vermicompost, Green manuring, Oilcakes).	2
	Total Theory classes	**32**

Practical

Introduction of analytical instruments and their principles, calibration and applications of Coloremetry and flame photometry; Estimation of soil organic carbon Estimation of alkaline hydrolysable N in soils; Estimation of soil extractable P in soils; Estimation of exchangeable K in soils; Estimation of exchangeable Ca and Mg in soils; Estimation of soil extractable S in soils; Estimation of DTPA extractable Zn in soils; Estimation of N in plants; Estimation of P in plants; Estimation of K in plants; Estimation of S in plants.

Practical Schedule

SN	Topic	No. of lectures
1.	Introduction of analytical instruments and their principles, calibration and applications of Coloremetry and flame photometry	2
2.	Estimation of organic carbon in soil	1
3.	Estimation of alkaline hydrolysable N in soils	1
4.	Estimation of soil extractable P in soils	2
5.	Estimation of exchangeable K in soils	1
6.	Estimation of exchangeable Ca and Mg in soils;	1

7.	Estimation of soil extractable S in soils	1
9.	Estimation of DTPA extractable Zn in soils	1
10.	Estimation of N in plants	2
11.	Estimation of P in plants	2
12.	Estimation of K in plants	1
13.	Estimation of S in plants	1
	Total practical classes	**16**

Suggested readings

1. Introductory Soil Science by Dilip Kumar Das, Kalyani Publishers
2. Soil Fertility and Nutrient Management by S. S. Singh, Kalyani Publishers
3. Soil Fertility and Fertilizers by Samual L. Tisdale, Werner L. Nelson and James D. Beaton, Macmillan Publishing Company, New York
4. The nature and Properties of Soils by Harry O. Buckman and Nyle C.

NCC-II NATIONAL CADET CORPS –II 1(0+1)

1. Values and ethics, perception, communication, motivation, decision making, discipline and duties of good citizen.
2. Leadership traits, types of leadership. Character/personality development.
3. Civildefenseorganization,typesofemergencies,firefighting,protection,
4. Maintenance of essential services, disaster management, aid during development projects.
5. Basics of social service, weaker sections of society and their needs, NGO's and their contribution, contribution of youth towards social welfare and family planning.
6. Structure and function of human body, diet and exercise, hygiene and sanitation.
7. Preventable diseases including AIDS, safe blood donation, first aid, physical and mental health.
8. Adventure activities
9. Basic principles of ecology, environmental conservation, pollution and its control.
10. Precaution and general behaviour of girl cadets, prevention of untoward incidents, vulnerable parts of the body, self defense.

NSS-II NATIONAL SERVICE SCHEME-II 1(0+1)

1. Importance and role of youth leadership: Meaning, types and traits of leadership, qualities of good leaders; importance and roles of youth leadership

2. Life competencies: Definition and importance of life competencies, problem-solving and decision-making, inter personal communication
3. Youth development programmes: Development of youth programmes and policy at the national level, state level and voluntary sector; youth-focused and youth-led organizations.
4. Health, hygiene and sanitation: Definition needs and scope of health education; role of food, nutrition, safe drinking water, water born diseases and sanitation (Swachh Bharat Abhiyan) for health; national health programmes and reproductive health.
5. Youth health, lifestyle, HIVAIDS and first aid: Healthy life styles, HIVAIDS, drugs and substance abuse, home nursing and first aid Youth and yoga
6. History, philosophy, concept, my thsandmis conceptions about yoga; yoga traditions and its impacts, yoga as a tool for healthy lifestyle, preventive and curative method.

PHED-II PHYSICAL EDUCATION AND YOGA PRACTICES 1(0+1)

1. Teaching of skills of Ball Badminton–involvement of all the skills in game situation with teaching of rule of the game
2. Teaching of some of Asanas –demonstration, practice, correction and practice
3. Teaching of some more of Asanas –demonstration, practice, correction and practice
4. Teaching of skills of Table Tennis – demonstration, practice of skills, correction and practice and involvement in game situation
5. Teaching of skills of Table Tennis – demonstration, practice of skills, correction and practice and involvement in game situation
6. Teaching of skills of Table Tennis–involvement of all the skills in game situation with teaching of rule of the game
7. Teaching–Meaning, Scope and importance of Physical Education
8. Teaching–Definition, Type of Tournaments
9. Teaching–Physical Fitness and Health Education
10. Construction and laying out of the track and field (*The girls will have Tennikoit and Throw Ball).